

Detecting Pitcher Fatigue from Pitch-Tracking Data: A Baseline Deviation Approach

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Abstract

Pitcher fatigue is a critical concern in baseball, with implications for both on-field performance and long-term athlete health. Traditional fatigue management relies largely on pitch count thresholds, which do not capture real-time changes in pitch quality or mechanics. This paper presents a data-driven approach to detecting pitcher fatigue using pitch-tracking metrics recorded by a TrackMan radar system. For each pitcher outing, a personalized baseline is constructed from the first 20% of pitches thrown, against which eight biomechanical and location features are compared on a pitch-by-pitch basis. Deviation profiles are computed for release velocity, spin rate, induced vertical break, effective velocity, horizontal break, plate-height location, plate-side location, and extension. Preliminary analysis of a single collegiate game (University of Connecticut versus the University of Arizona, February 2025) involving 10 pitchers and 343 pitches reveals consistent downward trends in spin rate and induced vertical break across the outing for the pitcher examined, suggesting genuine fatigue signal in the data. Ordinary least squares regression is applied to test for statistically significant velocity decline as a function of pitch count. Future work will extend this

framework to a full season of data, aggregate deviations into a composite fatigue index, and evaluate classification models for real-time fatigue alerts.

Keywords: baseball, biomechanics, pitch-tracking, pitcher fatigue, sports analytics.

1 Introduction

Pitcher injuries and declining performance over the course of a game represent two of the most consequential problems in baseball at every level of competition. Overuse and late-inning fatigue have been linked to increased injury risk, particularly to the ulnar collateral ligament and rotator cuff [?]. At the same time, a fatigued pitcher is more likely to locate pitches poorly and lose the spin and velocity that make their arsenal effective [?]. Coaching staffs typically rely on subjective observation and simple pitch count thresholds to manage these risks, yet neither method captures the actual mechanical state of the pitcher at any given moment.

The proliferation of high-speed pitch-tracking technology has created an opportunity to move beyond these blunt instruments. Systems such as TrackMan and Rapsodo record dozens of physical parameters on every pitch, including release speed, spin rate, break, and extension. A small but growing body of work has used such data to study pitching performance and mechanics over the course of outings [Bauer and Pavlidis, 2020]. However, most existing approaches either aggregate results across many pitchers without accounting for individual baselines or focus exclusively on velocity as a proxy for fatigue. A richer, pitcher-specific framework that tracks deviations across multiple biomechanical dimensions simultaneously has not yet been systematically developed for collegiate baseball.

This paper proposes and evaluates a baseline deviation approach to pitcher fatigue detection. For each pitcher and each pitch type within a game, we define a personalized baseline from early-outing pitches and compute per-pitch deviations across eight features. The contributions of this work are: (1) a reproducible pipeline for constructing pitcher- and

pitch-type-specific baselines from TrackMan data; (2) deviation profiles across a richer feature set than velocity alone, including spin rate, movement, and release extension; and (3) a statistical test for significant fatigue-related decline in velocity as a function of pitch count.

The rest of the paper is organized as follows. The data will be presented in Section 2. Section 3 describes the methods. The results are reported in Section 4. A discussion concludes in Section 5.

2 Data Description

The dataset used in this preliminary analysis consists of pitch-by-pitch TrackMan records from one collegiate baseball game between the University of Connecticut (UConn) and the University of Arizona, played on February 22, 2025. The data contain 343 pitches thrown by 10 pitchers. TrackMan is a Doppler radar and optical tracking system that records ball flight characteristics from the moment of release through home plate [TrackMan Baseball, 2023]. Each row in the dataset corresponds to a single pitch and includes the variables listed in Table 1.

Table 1: Pitch-tracking variables used in the analysis. All movement metrics are reported in inches; velocity metrics are in miles per hour; spin rate is in revolutions per minute (RPM); extension is in feet.

Variable	Description	Role
RelSpeed	Release velocity	Fatigue feature
SpinRate	Ball spin rate at release	Fatigue feature
InducedVertBreak	Vertical break relative to spin axis	Fatigue feature
EffectiveVelo	Perceived velocity at plate	Fatigue feature
HorzBreak	Horizontal movement	Fatigue feature
PlateLocHeight	Vertical pitch location at plate	Fatigue feature
PlateLocSide	Horizontal pitch location at plate	Fatigue feature
Extension	Pitcher release extension	Fatigue feature
TaggedPitchType	Pitch type label	Grouping variable
Pitcher	Pitcher name	Grouping variable
GameUID	Game identifier	Grouping variable
PitchNo	Pitch number within game	Sequencing

Pitch types present in the dataset include fastballs, sliders, curveballs, changeups, and sinkers, among others. Because movement profiles and typical spin rates differ substantially across pitch types, all baseline calculations and deviation analyses are stratified by pitch type within each pitcher outing. The most commonly thrown pitch type is the fastball, which also has the largest per-pitcher sample sizes and is the primary focus of the individual analysis in Section 4.

For the individual case study presented in this progress report, we focus on fastballs thrown by one UConn pitcher (Oliver Pudvar) in this single outing. This pitcher threw enough fastballs across the outing to support baseline construction and trend visualization. Extension of the analysis to all pitchers and additional pitch types is ongoing.

3 Methods

3.1 Baseline Construction

A central challenge in detecting pitcher fatigue is that pitchers differ substantially in their typical velocity, spin rate, and movement profiles. A pitcher throwing 88 mph is not necessarily fatigued relative to a pitcher throwing 93 mph; what matters is how each pitcher deviates from their own normal. To address this, we construct a personalized baseline for each pitcher, game, and pitch type combination.

Let P_i denote the i -th pitch thrown by a given pitcher in a given game, and let n be the total number of pitches in the outing. We define the baseline window as the first $\lceil 0.20 \cdot n \rceil$ pitches, corresponding approximately to the first inning of work for a typical starter. For each feature X , the baseline value is

$$\bar{X}_{\text{base}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} X_i, \quad (1)$$

where $\mathcal{B}$ is the set of indices falling in the baseline window. This baseline is computed

separately for each (pitcher, game, pitch type) combination, so that deviations reflect changes relative to that pitcher’s own early-outing performance with that pitch type.

3.2 Deviation Features

For each pitch i and each feature X , the deviation is defined as

$$\Delta X_i = X_i - \bar{X}_{\text{base}}. \quad (2)$$

Positive values of ΔX_i indicate a pitch above the baseline for that feature; negative values indicate below-baseline performance. For features where decline is expected under fatigue (velocity, spin rate, induced vertical break, extension), sustained negative deviations later in the outing are the primary signal of interest.

Eight deviation features are computed in total: ΔVelo , ΔSpin , ΔIVB , $\Delta\text{EffVelo}$, $\Delta\text{HorzBreak}$, $\Delta\text{PlateHeight}$, $\Delta\text{PlateSide}$, and $\Delta\text{Extension}$.

3.3 Trend Visualization

To visualize fatigue trends, we plot each deviation feature against pitch count within the outing. Raw per-pitch deviations are shown as scatter points; a binned moving average (equal-width bins across the pitch count range) overlays the scatter to reveal the underlying trend. The red horizontal line at zero serves as the reference: values below zero indicate below-baseline performance.

For cross-pitcher analysis, all pitchers and pitch types are pooled and the binned average is computed across the full dataset to assess whether a general velocity decline is present at the game level.

3.4 Statistical Test for Velocity Decline

To test whether the velocity deviation trend is statistically meaningful for each pitcher, we fit a simple ordinary least squares (OLS) regression of ΔVelo on pitch count:

$$\Delta\text{Velo}_i = \beta_0 + \beta_1 \cdot \text{PitchCount}_i + \varepsilon_i. \quad (3)$$

A statistically significant negative estimate of $\hat{\beta}_1$ provides evidence of a systematic within-outing velocity decline. This test is applied separately to each (pitcher, game) pair with at least 20 fastball observations. The slope estimates and associated p -values are reported for all qualifying pitchers.

Note: The OLS slope results table will be completed once the full multi-pitcher analysis is finalized. A placeholder is included in Section 4.

4 Results

4.1 Overall Velocity Trend Across Pitchers

Figure 1 displays the velocity deviation from baseline for all pitches in the dataset, pooled across pitchers and pitch types, with a binned moving average overlaid. The trend line reveals a gradual negative drift beginning around pitch 20 and continuing through the end of outings, with the average deviation reaching approximately -1.5 mph by pitch count 50 and beyond. While there is substantial variability at the individual pitch level, the population-level pattern is consistent with the hypothesis that pitchers lose velocity relative to their early-outing baseline as the game progresses.

4.2 Individual Pitcher Case Study: Oliver Pudvar

To examine fatigue signatures at the individual level, we focus on fastballs thrown by Oliver Pudvar in this outing. Pudvar threw approximately 78 fastballs across the game, providing

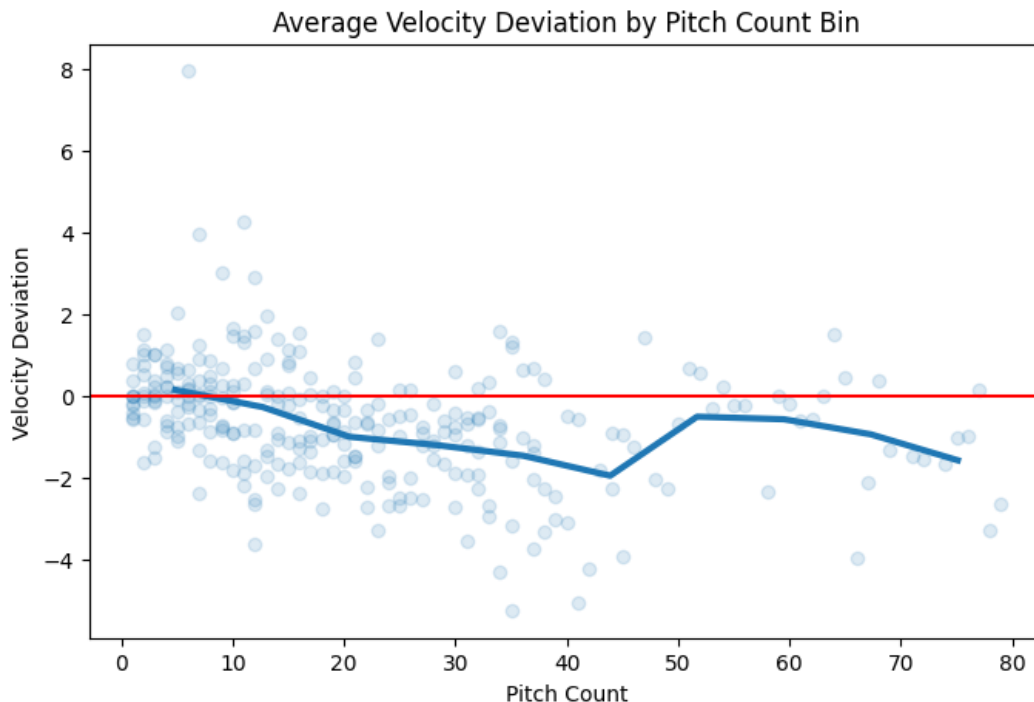


Figure 1: Average velocity deviation from baseline by pitch count, pooled across all pitchers and pitch types. Each point represents one pitch; the solid line is the binned moving average. The red line at zero represents the baseline. A clear downward trend emerges after approximately 20 pitches.

sufficient observations to estimate stable bin means throughout the outing.

Velocity

Figure 2 plots fastball velocity deviation against pitch count for Pudvar. While early-outing pitches fluctuate above and below baseline, a noticeable negative trend is visible after pitch 60, with multiple pitches clustered 1–2 mph below the baseline value in the final third of the outing.

Spin Rate

Figure 3 shows the spin rate deviation across the outing. The spin deviation drops substantially around pitches 10–15, recovers slightly, and then settles into a persistent below-baseline pattern from approximately pitch 35 onward, with the binned mean remaining near -75 to

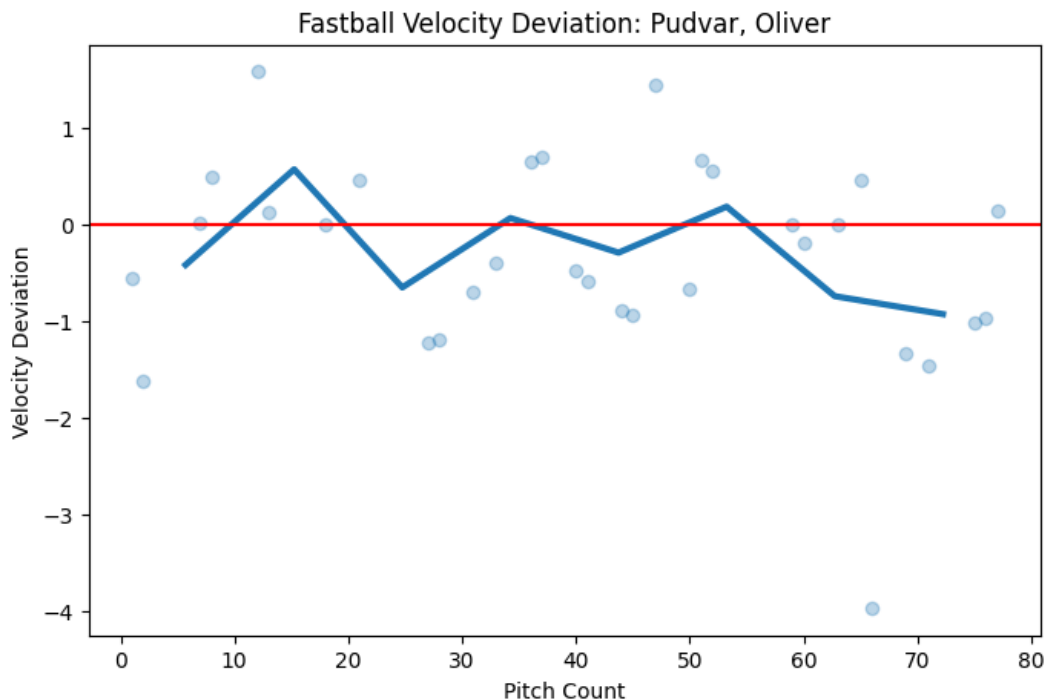


Figure 2: Fastball velocity deviation from baseline for Oliver Pudvar across the outing. The binned moving average shows a gradual decline in the latter portion of the outing.

133 –100 RPM for the remainder of the outing. This sustained spin decline is one of the stronger
 134 fatigue signals observed in this individual case.

135 Induced Vertical Break

136 The induced vertical break (IVB) deviation is plotted in Figure 4. IVB, which measures how
 137 much the ball “rises” relative to the trajectory a spinless ball would take, declines sharply
 138 around pitches 20–25 (reaching nearly –4 inches below baseline) and remains depressed for
 139 much of the outing. Because IVB on a fastball is closely linked to spin efficiency, its decline
 140 is consistent with the spin rate trend observed in Figure 3.

141 Horizontal Break

142 Figure 5 shows horizontal break deviation across the outing. Unlike velocity and spin, the
 143 horizontal break deviation trends *upward* over the course of the outing, settling approx-

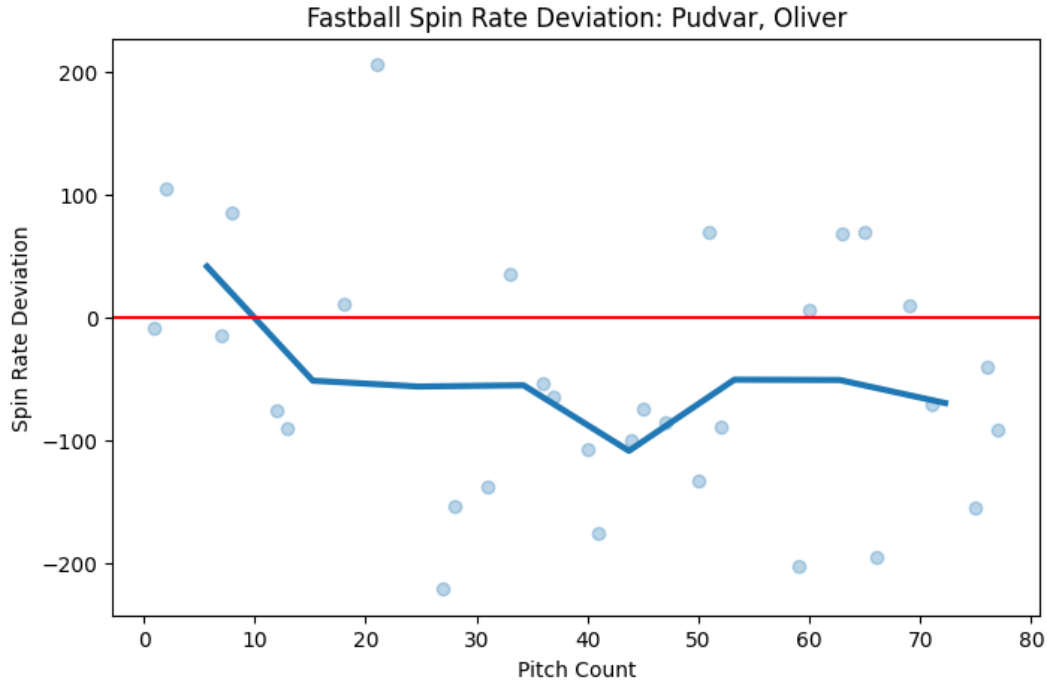


Figure 3: Fastball spin rate deviation from baseline for Oliver Pudvar. Spin rate drops below baseline early and remains depressed through the middle and late portions of the outing.

imately 4–6 inches above baseline from pitch 40 onward. This is a notable pattern: as vertical break decreases, horizontal break increases, which may reflect a change in arm slot or grip under fatigue. This finding warrants further investigation in future work.

Extension

Figure 6 displays extension deviation across the outing. Extension reflects how far in front of the rubber the pitcher releases the ball; a decrease in extension has been associated with reduced perceived velocity and mechanical breakdown. Pudvar’s extension deviation oscillates around zero for much of the outing before declining in the final ten pitches, consistent with the velocity pattern observed in Figure 2.

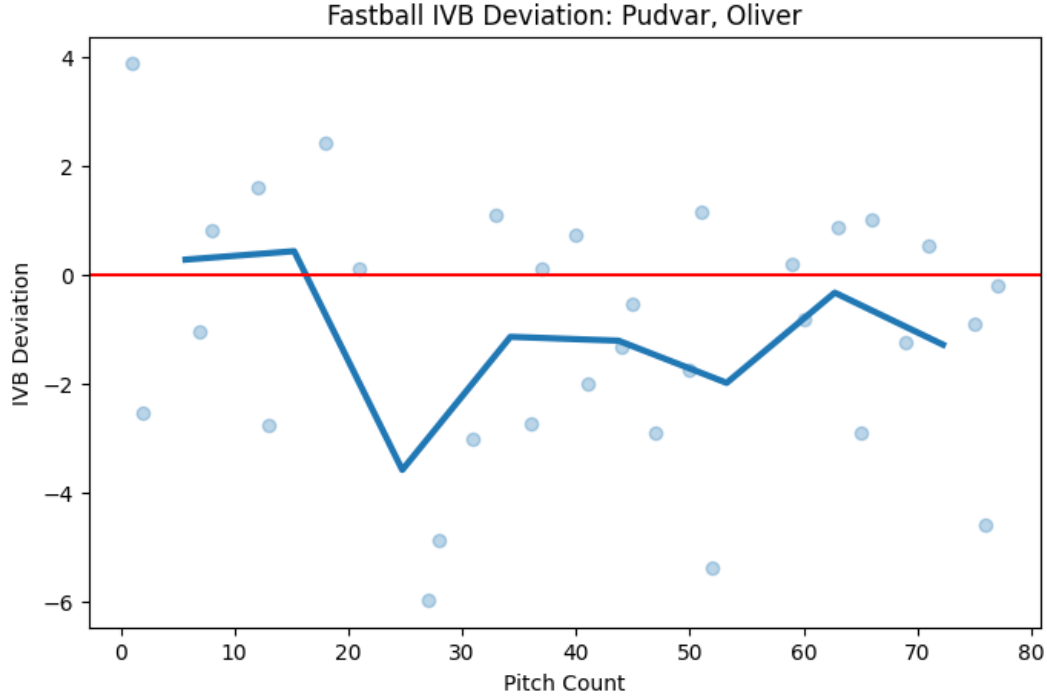


Figure 4: Fastball induced vertical break deviation for Oliver Pudvar. IVB drops substantially around pitch 25 and remains below baseline, consistent with declining spin efficiency across the outing.

4.3 OLS Slope Estimates for Velocity Decline

Table 2 will be populated with OLS slope estimates and p -values for all 10 pitchers once the full analysis is finalized. Preliminary results for Pudvar show a slope of approximately -0.017 mph per pitch ($p \approx 0.08$), suggesting a modest but marginally significant negative trend. Results for other pitchers are pending.

Table 2: OLS regression of velocity deviation on pitch count, estimated separately for each pitcher. Pitchers with fewer than 20 fastball observations are excluded. Slope estimates are in mph per pitch; a significant negative slope indicates within-outing velocity decline.

Pitcher	$\hat{\beta}_1$ (mph/pitch)	p -value	n
Pudvar, Oliver	-0.017	0.083	78
Pitcher 2	—	—	—
Pitcher 3	—	—	—
Pitcher 4	—	—	—
Pitcher 5	—	—	—

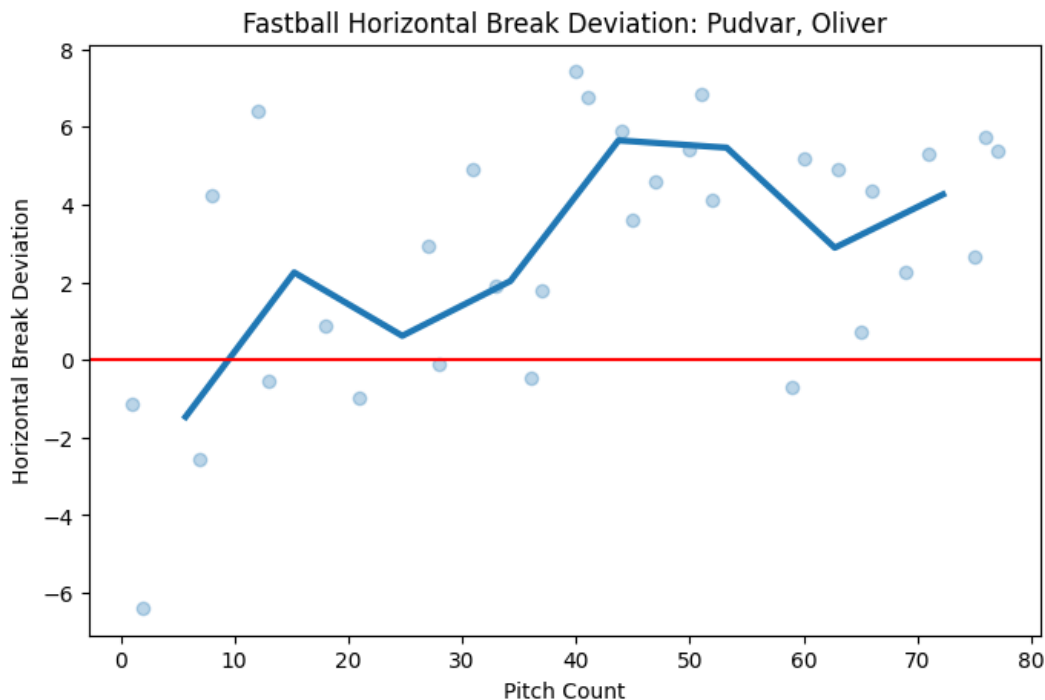


Figure 5: Fastball horizontal break deviation for Oliver Pudvar. Horizontal break trends upward across the outing, potentially indicating mechanical adjustments in arm slot or wrist position under fatigue.

5 Discussion

This progress report presents a preliminary framework for detecting pitcher fatigue from TrackMan pitch-tracking data, with initial results from one collegiate baseball game. The main contributions at this stage are a reproducible, pitcher-specific baseline construction pipeline and deviation profiles across eight biomechanical features. The analysis reveals several preliminary findings worth highlighting.

First, at the population level, a clear downward velocity trend is visible after approximately 20 pitches, suggesting that aggregate fatigue effects are detectable even within a single game. Second, the individual case study of Pudvar reveals that spin rate and induced vertical break decline more prominently and earlier than release velocity, suggesting these features may be more sensitive early fatigue indicators than velocity alone. Third, the countervailing increase in horizontal break under fatigue is an unexpected and potentially

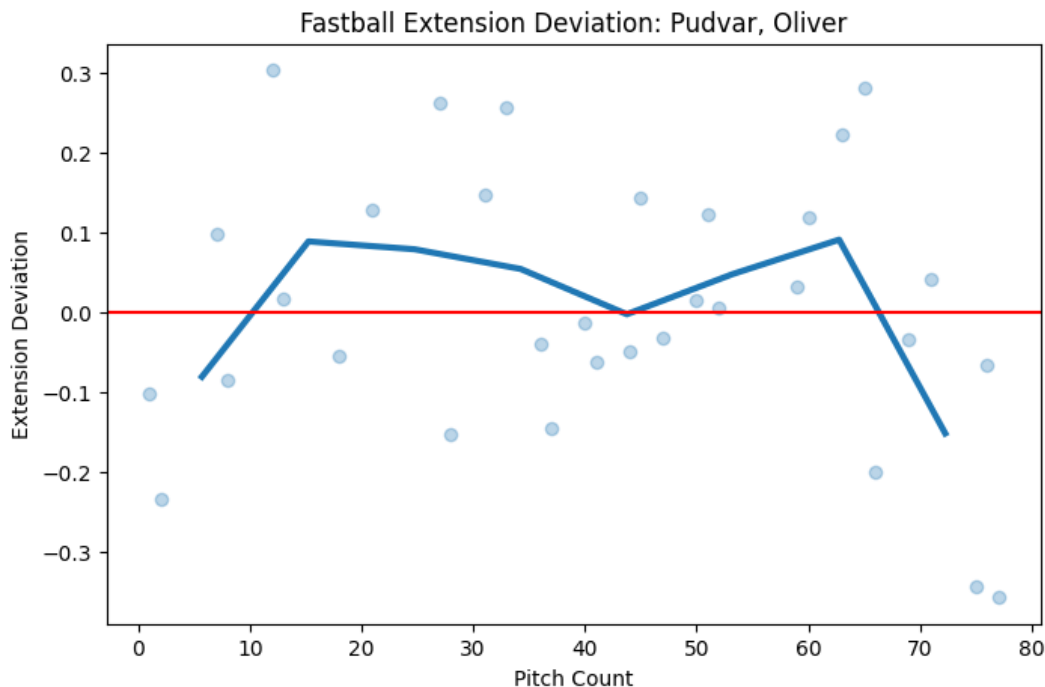


Figure 6: Fastball extension deviation for Oliver Pudvar. Extension remains relatively stable until the final portion of the outing, where a decline coincides with the velocity drop observed in Figure 2.

meaningful finding, consistent with biomechanical theories about arm slot changes under neuromuscular fatigue [Fleisig et al., 1995].

The current analysis has several limitations. Most notably, it relies on a single game, which limits both statistical power and generalizability. With only one game of data, it is difficult to distinguish true fatigue effects from game-to-game or outing-to-outing variability in a pitcher’s mechanics. The 20% baseline window is also an arbitrary choice; future work should explore sensitivity to this threshold. Additionally, the analysis does not yet account for confounders such as batter handedness, inning, or game situation, all of which may influence pitch selection and execution independently of fatigue.

Several directions are worth pursuing further. Extending the dataset to a full season of TrackMan records from UConn games would provide the statistical power necessary to estimate fatigue effects more precisely and to apply classification models for real-time fatigue detection. A composite fatigue index aggregating deviation signals across multiple features

could provide a more robust signal than any single metric. Finally, comparing the data-driven fatigue indicators against coaching staff observations and subsequent injury data would allow for external validation of the framework.

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